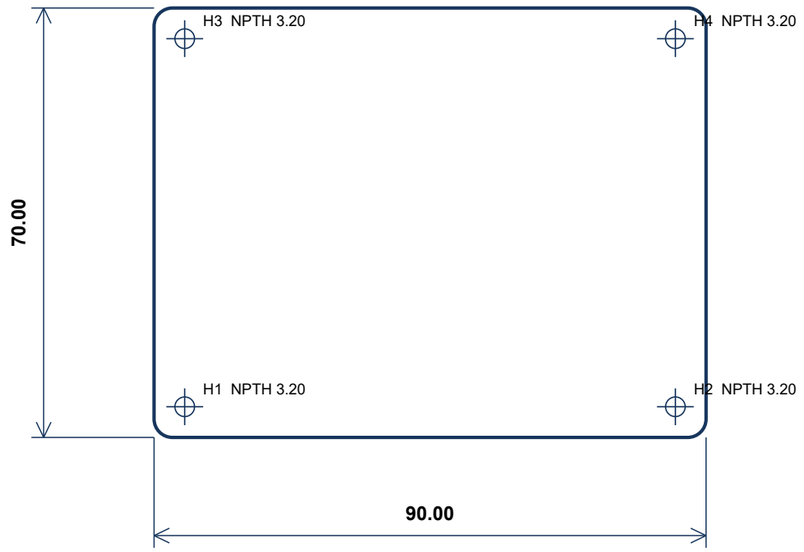
EVT_SYS_CARRIER REV.A | BOARD OUTLINE AND PILOT FABRICATION BASIS

Dimensions in millimetres. Mechanical datum and output status are bound to the final native Altium PcbDoc.

OUTLINE: R3.00 CORNERS, ROUTED PROFILE



MOUNTING DATUM: H1 CENTRE AT X=5.00, Y=5.00 FROM LOWER-LEFT BOARD EXTENTS

PILOT FABRICATION NOTES

1. Finished board outline: 90.00 x 70.00 mm with R3.00 corners.
2. Project mechanical target: 1.60 mm. Working Rezonit baseline: published 1.50 mm TG150 four-layer construction; order data prevails.
3. Four copper layers, 35 um base copper on every layer, FR-4 TG150 IPC-4101/99 working material.
4. L1 components/signals; L2 solid GND; L3 power and slow signals; L4 secondary signals.
5. Pilot finish request: ENIG, 0.075-0.125 um Au over 3-6 um Ni; green solder mask; white legend.
6. External interfaces are two 3.3 V UART channels and a connectorized external GNSS module. Ethernet is not fitted on this revision.
7. No controlled-impedance route class is released for this carrier revision. Any order-specific impedance request requires CAM solver review and a controlled PCB ECO.
8. Gerber X2 includes the routed board profile. Legacy RS-274X GKO is retained as a CAM fallback; NC drill files remain the drill authority.
9. Use the NC drill reports for plated, non-plated and Top-to-L2 tool definitions. Finished hole sizes are specified on sheet 2.
10. Electrical test is required. Submit single-board data; fabricator panelisation is permitted after CAM review.
11. IPC-2581 Revision B and Gerber X2 are both supplied. Resolve any CAM disagreement against the native PcbDoc and this drawing.
12. This pilot drawing supports quotation and CAM preflight. It does not grant production authority.

Item	Current evidence	Pilot R1 state
Gerber X2	4 copper, mask, paste, legend and profile	PASS
IPC-2581	Revision B; 83 components; current native export	PASS
Native DRC	0 warnings, 0 violations, 0 unrouted	PASS
Connections	483 pads, 1776 tracks, 241 vias, 60 fills	PASS
STEP	Board solid and hole geometry; component 3D out of scope	PASS SCOPE

EVT_SYS_CARRIER REV.A

PILOT R1 FABRICATION DRAWING
FABRICATOR REVIEW / CAM PREFLIGHT

Scale: 1:1

PRODUCTION AUTHORITY: NOT_AUTHORIZED

Sheet 1/2

EVT_SYS_CARRIER REV.A | STACKUP, SIGNAL-INTEGRITY, DFM AND DRILL SUMMARY

Pilot R1 uses the published Rezonit 4-layer, 35 um TG150 construction as a working baseline; order-specific values remain deferred.

PUBLISHED REZONIT WORKING STACK

Seq	Item	Layer / material	Thickness	Copper	Dk at 1 MHz	Status
1	Copper	L1 Top	0.035 mm	35 um		WORKING
2	Prepreg	FR-4 TG150 2116	0.115 mm		4.2	WORKING
3	Copper	L2 solid GND	0.035 mm	35 um		WORKING
4	Core	FR-4 TG150 IPC-4101/99	1.130 mm		4.6	WORKING
5	Copper	L3 power / slow signal	0.035 mm	35 um		WORKING
6	Prepreg	FR-4 TG150 2116	0.115 mm		4.2	WORKING
7	Copper	L4 Bottom	0.035 mm	35 um		WORKING

Published actual board thickness: 1.50 mm, excluding plating, finish and solder mask. Project mechanical target remains 1.60 mm.

SIGNAL-INTEGRITY DISPOSITION

Interface	Physical layer	Current revision	Fabricator action	Disposition
HOST_UART_A/B	3.3 V single-ended	Two connectorized UART channels	Check ordinary clearance and return path	NO Z0 CONTROL
GNSS	UART/PPS to external module	No onboard GNSS RF route	Check connector pinout and ESD placement	NO RF Z0 CLASS
Ethernet	Not fitted	Removed from canonical requirement	Do not add MDI artwork or impedance notes	NOT APPLICABLE

The published Rezonit construction is a working material baseline, not an order-bound impedance solution. The supplied artwork remains the validated EVT baseline. Any stack or routing change requires a controlled PCB ECO and repeated DRC and connection audit.

NC DRILL TOOL SUMMARY

Tool	Finished hole	Type	Count
T1	0.18 mm	PTH Top-L2	1
T2	0.20 mm	PTH through	133
T2	0.20 mm	PTH Top-L2	2
T3	0.25 mm	PTH	4
T4	0.30 mm	PTH	103
T5	0.40 mm	PTH	2
T6	0.991 mm	PTH	3
T7	1.02 mm	PTH	16
T8	3.00 mm	PTH	3
T9	3.00 mm	NPTH	1
T10	3.20 mm	NPTH	4
	TOTAL		272

PUBLISHED DFM BASELINE VS CURRENT DESIGN

Parameter	Current design / output	Rezonit published	Pilot disposition
Trace width	0.15 mm minimum rule	0.20 basic; 0.15 advanced	ADVANCED
Copper spacing	0.20 mm minimum rule	0.20 basic; 0.15 advanced	PASS BASIC
Drill	0.18 mm Top-L2; 0.20 mm through minimum	Order capability to be confirmed	CAM REVIEW
Via annular ring	Validate against current drill and pad tables	0.15 basic; 0.10 advanced	CAM REVIEW
NPTH	3.20 mm mounting holes	0.50 mm published minimum	PASS
Solder mask	Green; 0.025 mm expansion request	0.10 mm bridge; 0.025 expansion	CAM CHECK

SOURCE BINDING

Native PcbDoc SHA-256: 156db08f27896cf217806c3313631991c538db50ff1fdb9a1660058376a2e8ea
Native DRC: PASS, 0 warnings, 0 violations, 0 unrouted nets.
Connection integrity: PASS; all pads and routed copper remain inside the 90 x 70 mm outline.
Rezonit stack source: rezonit.ru/.../tipovye-sborki-i-konstruktsii-pechatnykh-plat/
Rezonit finish/mask source: rezonit.ru published production capabilities table.
Order-specific stack, finished copper and field-solver geometry take precedence when available.

PRODUCTION AUTHORITY REMAINS NOT_AUTHORIZED

EVT_SYS_CARRIER REV.A
PILOT R1 FABRICATION DRAWING
FABRICATOR REVIEW / CAM PREFLIGHT

Scale: 1:1
PRODUCTION AUTHORITY: NOT_AUTHORIZED